AOS-Bench: Can LLM Agents Manage Dynamic Workloads?

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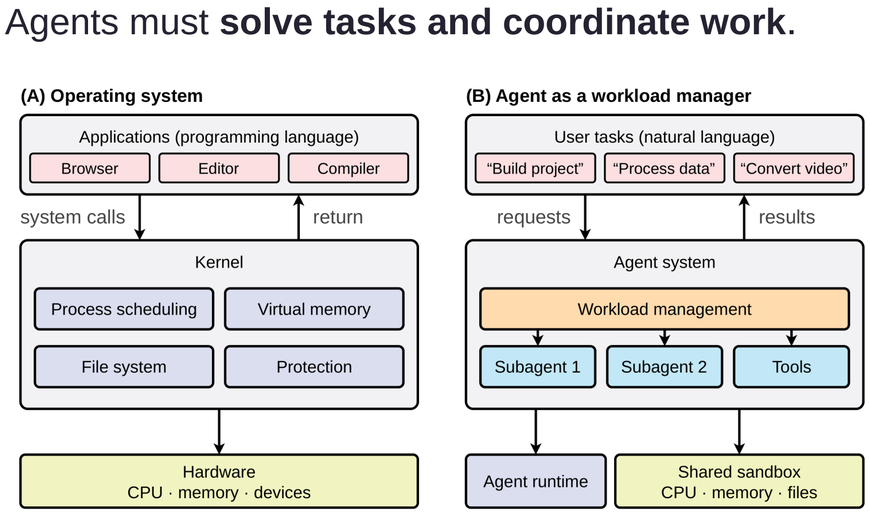
* Equal contribution † Corresponding author

Strong task solvers can still be weak workload managers.

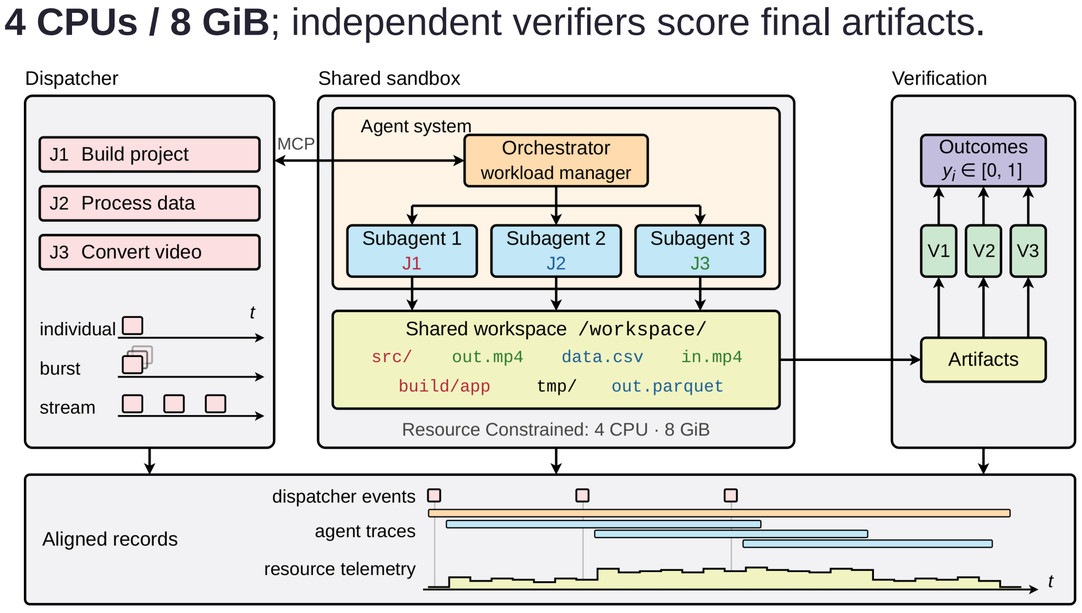
01 AOS-Bench: evaluate the agent as a workload manager

Requests arrive while other tasks are running.
Agents must **solve tasks and coordinate work**.

Individual: each task runs alone. **Stream:** tasks arrive over time.
Shared **4 CPUs / 8 GiB**; independent verifiers score final artifacts.



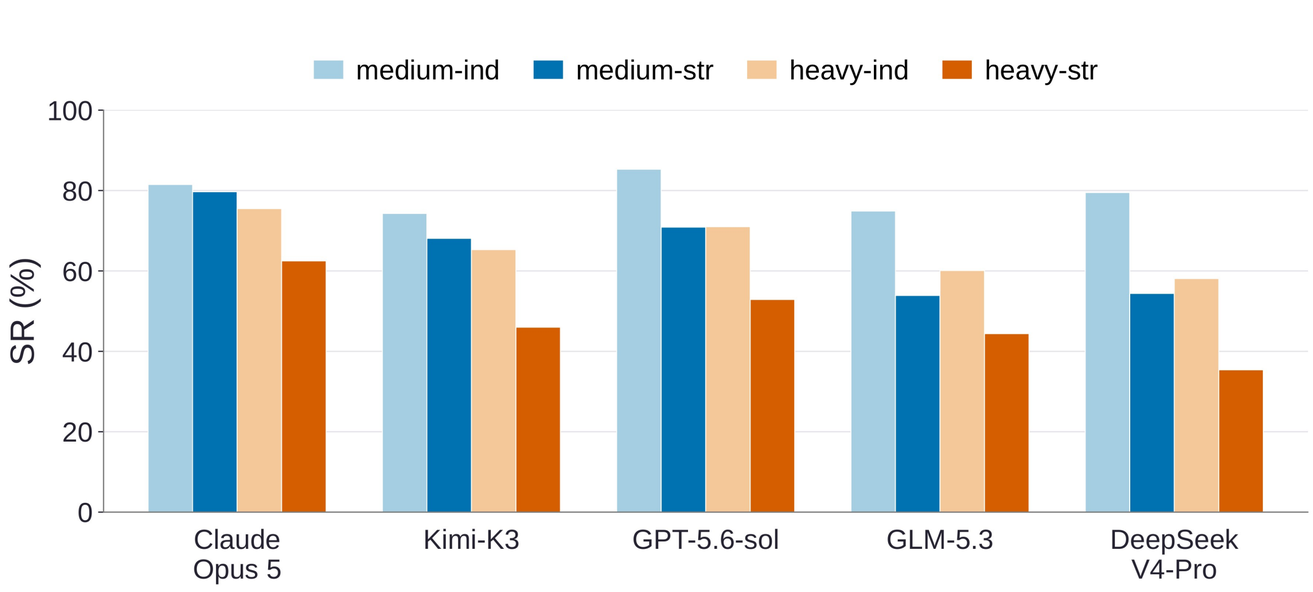
From isolated requests to persistent management of subagents, tools and shared resources.



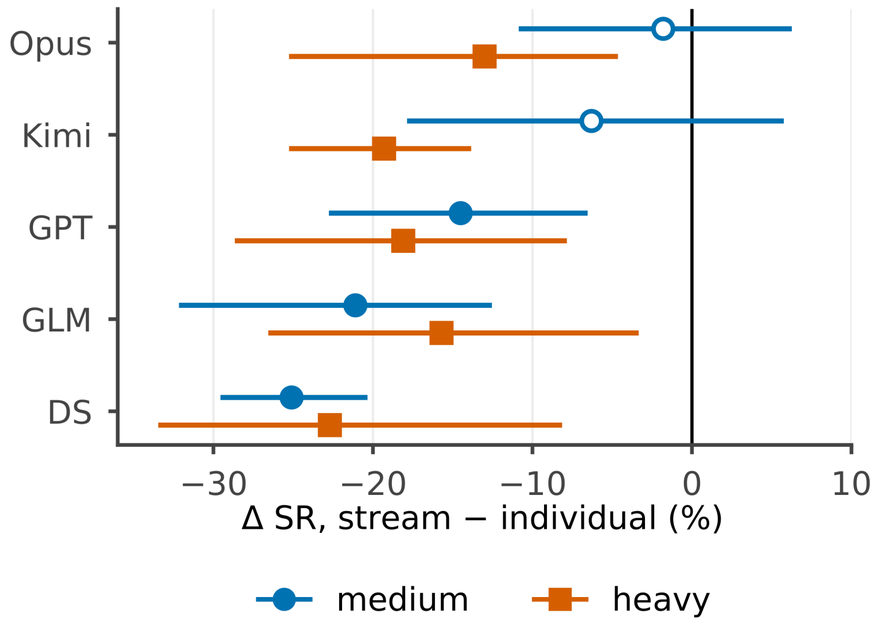
5 model–harness pairs • **Medium:** 10 tasks × 10 workloads • **Heavy:** 20 tasks × 5 workloads • Up to 3 repeats

02 Core result: isolated capability does not determine workload delivery

Every system loses **13.0–22.7 percentage points** on heavy workloads.



Paper SR values grouped by model; compare individual vs. stream within tier.



Original Fig. 5. Paired differences; 95% bootstrap intervals.

Medium: no system keeps its rank. GLM / Kimi: **74.9 / 74.3%** individually → **53.9 / 68.1%** in stream.

Faster delivery can coexist with lower SR: medium rewarded-task throughput is **1.49–2.22×** each system's sequential reference.

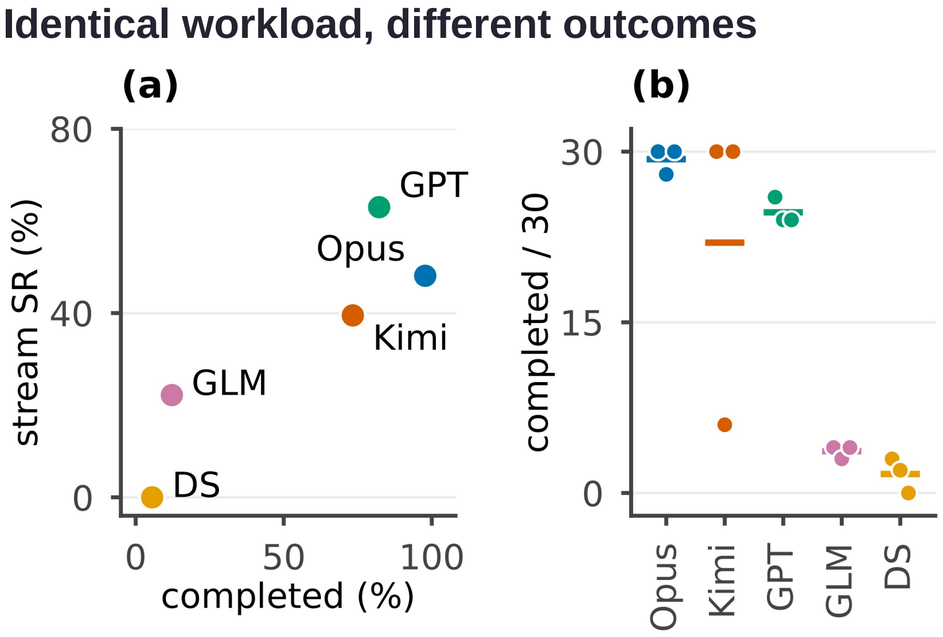
03 Why delivery falls: intake, concurrency and unstable execution

Controlled policy comparison · GLM-5.3

Tier	Policy	N	SR (%)	ET (min)	Claim (%)	Compl. (%)	t_{claim} (s)	CPU (min)
Med	Serial (k=1)	30/30	39.5	150.0	51.7	40.6	3360	98.5
Med	Native	30/30	53.9	150.1	75.9	60.9	1204	93.4
Med	Guidance	30/30	60.2	150.0	81.2	67.0	1642	134.0
Med	Fixed (k=2)	30/30	56.6	150.0	80.1	59.4	2614	181.1
Hvy	Serial (k=1)	15/15	28.6	240.0	30.0	23.6	4819	189.4
Hvy	Native	15/15	44.4	240.1	75.9	59.5	1190	268.7
Hvy	Guidance	15/15	39.2	240.1	86.1	63.7	2502	318.2
Hvy	Fixed (k=2)	15/15	42.8	240.0	53.6	43.5	3600	294.1

Original Table 3. Same solver, arrivals, tasks and resource budget.

Identical workload, different outcomes



Original Fig. 7. Thirty-task case study; three repeats per system.

73% never claimed
Serial FIFO on heavy workloads.

Fixed k = 2 is within 4 pp
Of the best managed policy in each tier.

Kimi completes 30 / 6 / 30
Serial collapse can be stochastic.

Scope: five systems; five heavy workloads; CLI tasks without dependencies. Heavy-tier intervals are a coarse check; the 30-task result is a case study.

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